Lesson 7 Date 15 No. 7

Model arxitekturasi tayyor bo'lgandan keyin Training jarayoni va u tez, arzon-roq va xatoliklarsiz bo'lishi juda muhim. Xatoliklar sabab model yaxshi natija bermay vaqt va pul yutqazamiz.
Training 3 ta asosiy qismdan iborat

① Hyperparameters – modelni qanday sharoitda qanday tezlikda o'qitishini belgilaydi. parameters ga learning rate, batch-size epochs va optimizer (AdamW) va boshqalar misol.

② Training stability – training jarayoni davomida model barqaror o'qitishi, xatoliklar bo'lmasligini oldini olib nazorat qilish kerak va jarayon to'xtab qolmasligi u-n harakat qilinadi.
Ex: Gradient clipping, LR schedule, Gradient vanishing/exploding, Loss nan yoki inf bo'lib qolishi.

③ Regularization – overfitting oldini olish yani yodlab olmasligi va ~~overfitting data~~ yangi ma'lumotlarda yaxshi ishlashi u-n yordam beradi. Lekin model LLM ya'ni ko'pincha datani yodlab olmaydi chunki tokenlar ko'pligi bois ulgurmaydi. Mashhur usullari → Weight decay va Dropout. Weight decay'ni foydalanish yetarli. Dropout ishlatilinmaydi.

Training hyperparameters
Hyperparameter model qanday o'qitishi, qancha vaqt train qilishi, xatolarni qanday kamaytirish va natijani yaxshilash uchun ta'sir qiladigan settingsga o'xshash.

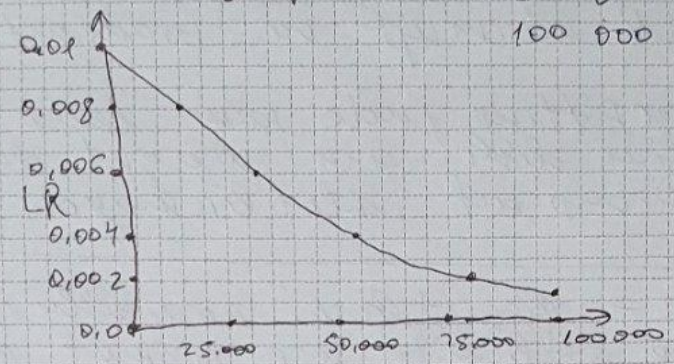
① Learning rate bu o'qitish qadami. Model loss'ni kamaytirish uchun xatoni qayta update qilib Gradient decent b-n

bilan hisoblaganda beriladigan qadam.
Batch-size - model 1ta trainda nechta training misolini qayta ishlashni belgilaydi.
Ex: 100 000 ta test sample bor batch-size 100 bersak 1000 ta batch bo'ladi.
Kichik batch-size bersak kamroq GPU xotira talab qilishi mumkin lekin bersak tezkorligi. Lekin kichik batch-size Gradient hisoblashda noisy kattaroq batch-size ga nisbatan barqarorroq.
Bu yerda Gradient Accumulation degan narsa bor yani GPU xotirasi bo'sa katta batch-size ni ko'tara olmay qolishi m-n.
Optimizer bu gradient bilan hisoblab model parameters ni yangi bosqichda qanday yangilanishni belgilab beradi.
Ex: AdamW. LLM da AdamW yaxshi sababi LLM da parameters juda ko'p va ularni gradient lari va qiymatlari turlicha bo'lishi mumkin. AdamW har bir parameter n-n moslashadi va update qilishga yordam beradi.
Warmup steps bu training boshida kichik qiymatdan boshlab kattalashtirib borish. ma'lum bir belgilab olgan qadamgacha.
Ex: Jami training step 100000 Warmup step 1000 LR: 0,001 500 - warmup stepda 0,0005 bo'ladi LR. 750 - warmup stepda 0,00075 1000 - stepda 0,001 bo'ladi.
O'zi nega warmup step kerak? sababi ~~boshida~~ training boshlab model parameters ga moslashishi kerak aks holda katta LR b-n boshlasak trainingda unstable bo'lishi m-m.
Weight Decay - bu modelning parameters juda katta bo'lib ketishi cheklaydi

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va generalization qobilyatini yaxshilashga yordam beradi.

LR scheduler bunda LR training davomida doimo bir xil qolishi shart emas. U vaqt yoki steps ga qarab o'zgaradi



Frontier modelsda cosine decay ishlatilishi yaxshi bunda LR cosinus bo'yicha kamaytiriladi. Cosine decay train oxirida LR ni deyarli nolga tushiradi. Training oxirroqida updates kichikroq bo'lgani yaxshi.

Frontier modelsda Global batch size 1M dan 4M gacha tokenni hisoblaydi. 1 ta stepda GPU qancha tokenni hisoblashi Global batch-size. Device ga qarab batch-size belgilanadi. GPU katta kichikligiga qarab 2, 4, 8 yoki 16.

Gradient Clipping - bu gradient qiymatlari juda katta bo'lib ketganda chegara qo'yib shu chegaradan oshirmaslik. Agar chegara qo'ymasa loss katta training bo'lgani. Ko'pincha Gradient Norm Clipping usuli foydalaniladi va Frontier models Max-norm ni 1 qilib tanlaydi.

Gradient Exploding bu LR katta berilsa muammo bo'ladi loss kattaradi / nolga teng bunda Loss = NaN yoki Loss = Inf bo'lib qoladi.

Gradient Exploding — layer kop bolsa
Warm up qilinmasa, xato matn lar
juda kop bolsa.
Kop layer nega yaxshi chunki backpropaga-
tionda u layerdan layerga gradient
kattalashib exploding bo'lishi m-m.
Yechim: LR warm up va Gradient
Clipping.
LLMda Overfitting juda kam lekin
baribir oldini olish u-n Weight decay
yaxshi. Droup out ishlatilmaydi

Normalization — qiymatlarni normallashtirib training stability u-n xizmat qiladi.
Post-norm eski yo'l Attention va MLP(FFN) dan keyin qo'llanilgan. Hozir Pre-Norm Attention va MLP dan oldin qo'llanilodi. Oldin norm qo'llangan training osonlashadi. Norm qilishning RMSNorm yaxshi. Chunki hisob kitob kam va GPU kam yetdi.
Bias term Zamonaviy modellarda Bias bo'lmasa parameters kamayadi → hisob kitob kamayadi → resurs tejaymiz.
MLP activations ReLU faqat musbat qismi u-n ishlatmaymiz. GeLU input qiymatga qarab o'rgatiradi. eski usullar bu ikkisi.
SwiGLU va GeGLU esa gating mexanizmidan foydalanadi, ya'ni ma'lum bir qismni boshqa qism bilan boshqaradi. Zamonaviy yo'l.
Embeddingda ROPE yaxshi Position va absolute usullar birlashmasidan yaratilgan ikkisining ham ishini qila oladi.
Attention inference eski usul MHA u 1 ta K va V kombinatsiyasi ko'proq ketdi. Keyinroq GQA zamonaviy tezroq

MHA MHA dan 4 marta kichrog tezrog
kamrog xotira.

Tokenization. Ġ - maxsus
I study hard
"I" "Ġstudy" "Ġhard"
Ġ - space borligini bildiradi,
I ni oldida space yog

LLM architecture old and modern

Project idea → resurses calculation → Data cleaning → Deduplication, filtering toxic and harmful language filter

→ Tokenization with BPE → Input Embedding → old: Absolute positional; now: RoPE

→ Normalization → Old: Post-Norm → MLP va Attention dan keyin oldida; now: Pre-Norm → RMS Norm

→ Attention, MLP Parallel foydalanish ← MHA old way; GQA now; old: ReLU GeLU; now: SwiGLU GeGLU

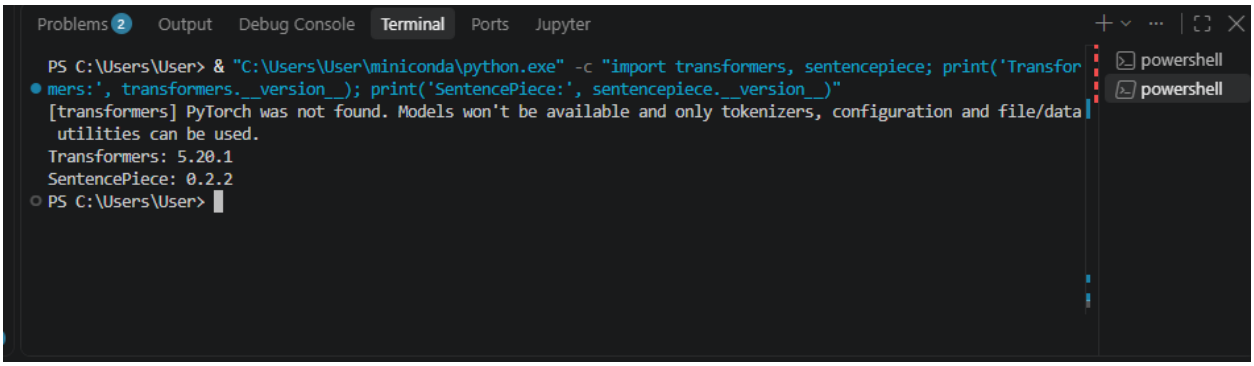
→ Pre-training → set model hyperparameters →

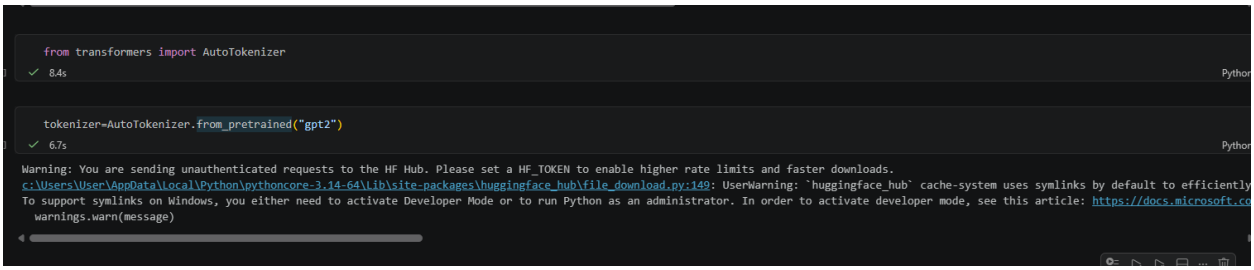
→ Unsupervised Pre-training → Fine-tuning and Alignment → Alignment and RLHF

→ Product deploy → Monitoring Evaluation → ~~no old way~~ stability output stability attention

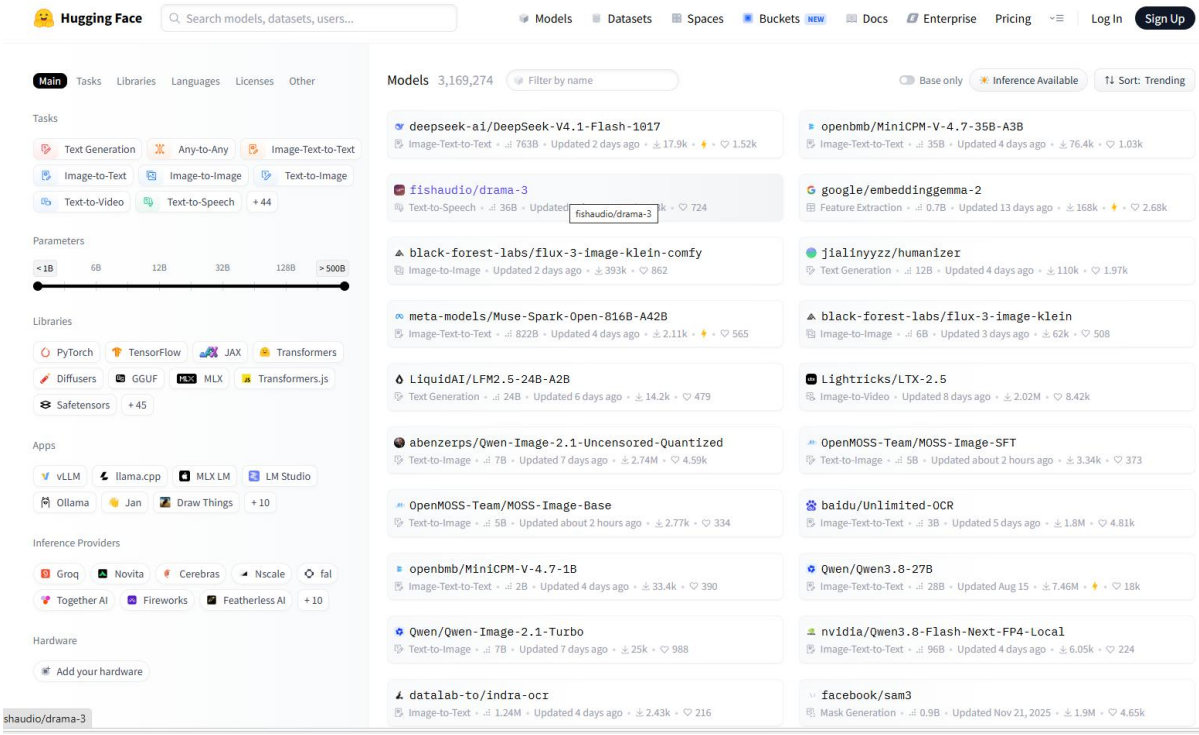
→ no old way and now: Z-loss

→ no old way and now: Q-K Norm (RMS Norm on Q and K)





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tokenizer_config	10/19/2026 12:33 PM	JSON Source File	1 KB
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Image-to-Text
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+ 44
Parameters
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6B
12B
32B
128B
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Apps
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MLX LM
LM Studio
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deepseek-ai/DeepSeek-V4.1-Flash-1017
Image-Text-to-Text • 763B • Updated 2 days ago • 17.9k • 1.52k
openbmb/MiniCPM-V-4.7-35B-A3B
Image-Text-to-Text • 35B • Updated 4 days ago • 76.4k • 1.03k
fishaudio/drama-3
Text-to-Speech • 36B • Updated
fishaudio/drama-3
k • 724
google/embeddinggemma-2
Feature Extraction • 0.7B • Updated 13 days ago • 168k • 2.68k
black-forest-labs/flux-3-image-klein-comfy
Image-to-Image • Updated 2 days ago • 393k • 862
jialinyyzz/humanizer
Text Generation • 12B • Updated 4 days ago • 110k • 1.97k
meta-models/Muse-Spark-Open-816B-A42B
Image-Text-to-Text • 822B • Updated 4 days ago • 2.11k • 565
black-forest-labs/flux-3-image-klein
Image-to-Image • 6B • Updated 3 days ago • 62k • 508
LiquidAI/LFM2.5-24B-A2B
Text Generation • 24B • Updated 6 days ago • 14.2k • 479
Lightricks/LTX-2.5
Image-to-Video • Updated 8 days ago • 2.02M • 8.42k
abenzerps/Qwen-Image-2.1-Uncensored-Quantized
Text-to-Image • 7B • Updated 7 days ago • 2.74M • 4.59k
OpenMOSS-Team/MOSS-Image-SFT
Text-to-Image • 5B • Updated about 2 hours ago • 3.34k • 373
OpenMOSS-Team/MOSS-Image-Base
Text-to-Image • 5B • Updated about 2 hours ago • 2.77k • 334
baidu/Unlimited-OCR
Image-Text-to-Text • 3B • Updated 5 days ago • 1.8M • 4.81k
openbmb/MiniCPM-V-4.7-1B
Image-Text-to-Text • 2B • Updated 4 days ago • 33.4k • 390
Qwen/Qwen3.8-27B
Image-Text-to-Text • 28B • Updated Aug 15 • 7.46M • 18k
Qwen/Qwen-Image-2.1-Turbo
Text-to-Image • 7B • Updated 7 days ago • 25k • 988
nvidia/Qwen3.8-Flash-Next-FP4-Local
Image-Text-to-Text • 96B • Updated 4 days ago • 6.05k • 224
datalab-to/indra-ocr
Image-to-Text • 1.24M • Updated 4 days ago • 2.43k • 216
facebook/sam3
Mask Generation • 0.9B • Updated Nov 21, 2025 • 1.9M • 4.65k
shaudio/drama-3